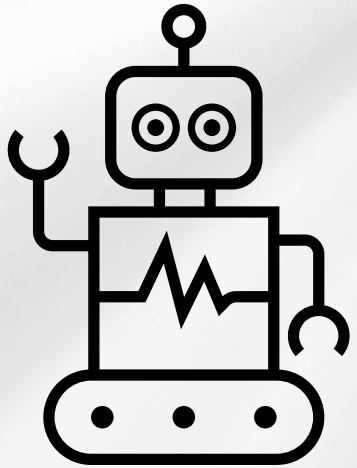
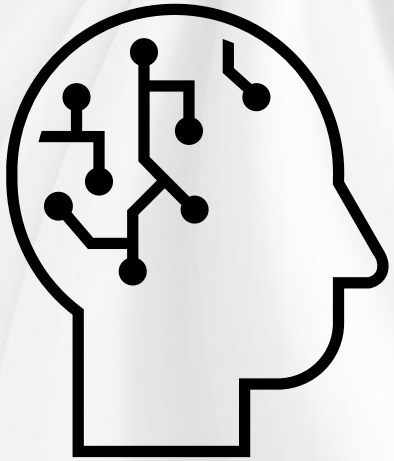
Introduction to AI Ethics





Roadmap

Introduction

Ethical Considerations in AI Development and Deployment

Historical Context and Evolution of AI Ethics

Case Studies

Introduction

AI Ethics

- Explores the moral and ethical implications of developing and deploying AI systems
- Considers issues such as fairness, privacy, transparency, accountability, and safety
- Seeks to promote the responsible and beneficial use of AI in various application domains while minimizing potential harms
- Involves various stakeholders including computer scientists, ethicists, policymakers, and users of target applications

Societal Impacts of AI

- Transforming industries and business models
- Enhancing decision-making and problem-solving
- Improving accessibility and inclusion
- Addressing global challenges
- Raising concerns about privacy, security, and autonomy
- Exacerbating socioeconomic inequalities
- Shaping the future of work and education

Some Real-World Examples of AI Dilemma

- Algorithmic bias in hiring and lending decisions
- Privacy concerns of facial recognition technology
- Autonomous vehicles and moral decision-making
- AI in healthcare and patient privacy
- Deepfakes and the spread of misinformation
- Algorithmic decision-making in criminal justice
- Job displacement due to AI automation

Ethical Considerations in AI Development and Deployment

Main Ethical Principles in AI Development

- Fairness and non-discrimination
- Privacy and data protection
- Transparency and explainability
- Accountability and responsibility
- Safety and robustness
- Beneficence and non-maleficence

Main Ethical Frameworks for AI

- Utilitarian ethics: focuses on maximizing overall well-being and minimizing harm
- Deontological ethics: emphasizes adherence to moral rules and duties
- Virtue ethics: centers on developing moral character and virtues in AI development and use

Ethical Implications of AI Systems

- Bias and discrimination perpetuate existing societal biases
- Privacy and surveillance concerns infringe on personal privacy
- Job displacement and economic inequality exacerbate disparities
- Autonomous decision-making and accountability issues in critical domains
- Manipulation and misinformation spread through AI-generated content
- Dependency on AI leads to loss of human skills

Long-Term Considerations

- Long-term existential risks posed by advanced AI systems
- Ethical considerations in AI research and development
- Evolving ethical frameworks to address emerging AI technologies

Stakeholders in Ethical AI Practices

Development and implementation

- Developers and researchers
- Companies and organizations

Governance and oversight

- Governments and policymakers
- Ethics boards and advisory committees

Public engagement and education

- End-users and consumers
- Civil society organizations and advocacy groups
- Educational institutions

Historical Context and Evolution of AI Ethics

From Ancient Myths to Early Years of AI

- Early AI concepts rooted in ancient myths and legends
- The term “artificial intelligence” coined in 1956 Dartmouth Summer Research Project
- Early AI research, which focused on symbolic reasoning
  - Raised initial ethical concerns about machine decision-making
  - Questioned potential impact on human autonomy
- AI winter of 1970s and 1980s
  - Temporarily slowed ethical discussions

Ethical Challenges from AI Resurgence

- AI resurgence in 1990s and 2000s driven by machine learning
  - Reignited debates on ethical issues about AI systems
- Advances of deep learning and big data analytics in 2010s
  - New ethical challenges related to privacy and bias
  - Raised concerns about superhuman capabilities in certain domains

Contemporary AI Ethics Discussions

Contemporary discussions cover a wide range of ethical issues

- Additional issues include transparency and accountability
- Sparked discussions on long-term implications of artificial general intelligence (AGI) and artificial superintelligence (ASI)

Emerging Ethical Challenges

- Advances in GenAI raised new concerns, e.g., misinformation, deepfakes, potential manipulation of human behavior
- Autonomous systems aroused ethical debates, e.g., self-driving cars, military applications
- Growing concerns about societal impact, e.g., job displacement, economic inequality
- Long-term considerations related to AGI

Some Recent Events Shaping AI Ethics

- 2008 financial crisis
- High-profile data breaches and privacy scandals
- 2016 U.S. presidential election and investigations into social media manipulation
- Recent advancements in AI-generated content

https://www.rand.org/pubs/perspectives/PEA2679-1.html

Case Study: AI in Hiring Practices

Background

TechCorp, a leading software development company, has recently implemented an AI-driven recruitment tool to streamline its hiring process.

The tool uses machine learning algorithms to analyze resumes and rank candidates based on their qualifications, experiences, and skills.

The company aims to reduce bias in hiring and improve efficiency by automating the initial screening process.

Background (2)

However, after several months of use, TechCorp's HR department notices troubling patterns.

The AI system seems to favor candidates from certain universities and backgrounds, leading to a lack of diversity in the applicant pool.

Additionally, some qualified candidates are being overlooked due to the AI's reliance on specific keywords that do not accurately reflect their skills or experiences.

Key Issues

- Algorithmic bias: The AI system appears to perpetuate existing biases by favoring candidates who fit a particular profile, which could lead to discrimination against underrepresented groups.
- Transparency: Candidates have raised concerns about a lack of transparency regarding how their resumes are evaluated and the criteria used by the AI system.

Key Issues (2)

- Accountability: There is ambiguity regarding who is responsible for the decisions made by the AI tool – should the blame fall on the developers, the HR team, or the company as a whole?
- Impact on diversity: The tool's performance raises questions about the company's commitment to diversity and inclusion, as it could hinder the hiring of diverse talent.

Discussion Questions

- Bias and fairness: What are the potential sources of bias in AI-driven hiring tools, and how can TechCorp address these biases to ensure fair hiring practices?
- Transparency: Should TechCorp disclose the criteria and algorithms used by the AI system to candidates? What are the implications of transparency for both the candidates and the company?

Discussion Questions (2)

- Accountability: In the event that the AI system leads to discriminatory hiring practices, who should be held accountable – the AI developers, the HR department, or the company leadership? Why?
- Diversity and inclusion: How can TechCorp balance the efficiency gains provided by AI with its commitment to diversity and inclusion in the hiring process?

Discussion Questions (3)

- Alternative solutions: What alternative methods could TechCorp implement to enhance its hiring process while minimizing bias and ensuring a diverse applicant pool?
- Long-term implications: What long-term effects could reliance on AI in hiring have on company culture and employee morale, particularly regarding diversity and inclusion efforts?

Case Study: AI in Predictive Policing

Background

CitySafe, a metropolitan police department, has implemented an AI-driven predictive policing system designed to forecast crime hotspots and allocate resources more effectively.

The system analyzes historical crime data, demographic information, and social media activity to identify areas with a higher likelihood of criminal activity.

Background (2)

While the goal is to enhance public safety and optimize police presence, community members have raised concerns about the potential for racial profiling and the erosion of civil liberties.

Over time, reports indicate that the system disproportionately targets neighborhoods with higher minority populations, leading to increased police presence and tensions in these communities.

Key Issues

- Algorithmic bias: The predictive policing system may reinforce existing biases present in historical crime data, disproportionately affecting marginalized communities and leading to racial profiling.
- Transparency: There is a lack of clarity regarding how the AI algorithms function and the data sources used, raising questions about accountability and the rights of citizens.

Key Issues (2)

- Civil liberties: The increased police presence in certain neighborhoods raises concerns about the infringement of civil liberties, including the right to privacy and protection from undue surveillance.
- Community trust: The use of AI in policing can erode trust between law enforcement and the communities they serve, particularly if residents feel they are being unfairly targeted.

Discussion Questions

- Bias and fairness: What strategies can CitySafe implement to identify and mitigate biases in its predictive policing algorithms to ensure fair treatment of all communities?
- Transparency: Should CitySafe publicly disclose the algorithms and data sources used in its predictive policing system? What are the potential benefits and drawbacks of such transparency?

Discussion Questions (2)

- Accountability: In cases where the predictive policing system leads to negative outcomes, who should be held accountable – the developers of the AI, the police department, or the city officials? Why?
- Balancing safety and rights: How can CitySafe balance the goal of enhancing public safety with the need to protect civil liberties and maintain community trust?

Discussion Questions (3)

- Alternative approaches: What alternative approaches could CitySafe consider for crime prevention that do not rely on AI and may foster better community relations?
- Long-term societal impacts: What are the potential long-term societal impacts of using AI in policing, particularly regarding community relations, crime rates, and public perception of law enforcement?